8	**Rectangular cartesian coordinates**	**Notes**
	The distance between two points.	The distance d between two points (x_1, y_1) and (x_2, y_2) is given by $d^2 = (x_1 - x_2)^2 + (y_1 - y_2)^2$.
	The point dividing a line in a given ratio.	The coordinates of the point dividing the line joining (x_1, y_1) and (x_2, y_2) in the ratio $m : n$ are given by $\left(\frac{nx_1+mx_2}{m+n}, \frac{ny_1+my_2}{m+n}\right)$
	Gradient of a straight line joining two points.	
	The straight line and its equation.	The $y = mx + c$ and $y - y_1 = m(x - x_1)$ forms of the equation of a straight line are expected to be known. The interpretation of $ax + by = c$ as a straight line is expected to be known.
	The condition for two lines to be parallel or to be perpendicular.	